WHAT DRIVES MOVIE SUCCESS? AN ANALYSIS OF BUDGET, RUNTIME, AND POPULARITY

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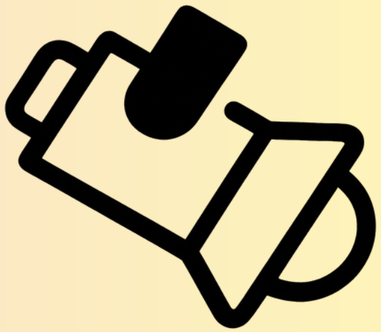
PROJECT OVERVIEW

Analyzing factors that influence movie performance



Why this topic?

- Interest in understanding what drives success in the film industry
- Availability of rich movie data
- Opportunity to apply machine learning techniques

Size & Scope

- Thousands of movies
- Multiple years & countries
- From low-budget to blockbusters

Features

- Budget & Revenue
- Runtime & Release Date
- Ratings (votes)
- Director, Genres, Companies

The Hypotheses

Budget

Hypothesis 1: Films with larger production budgets generate higher box office revenue compared to films with smaller budgets.

Director

Hypothesis 2: The choice of director significantly influences the production budget of a film.

Runtime

Hypothesis 3: Films with longer runtimes tend to receive lower popularity or audience approval compared to shorter films.

Data Preparation & Feature Engineering

Data Cleaning

- Handled missing values
- Corrected data types
- Removed unrealistic or inconsistent entries

Feature Engineering

- Created new variables to improve analysis
- Transformed data to handle skewness
- Extracted useful information from existing features

Feature Selection

- Selected relevant variables for modeling
- Removed redundant or noisy features

Modeling Setup

- Scaled features for consistency
- Split data into training and testing sets
- Ensured reliable model evaluation

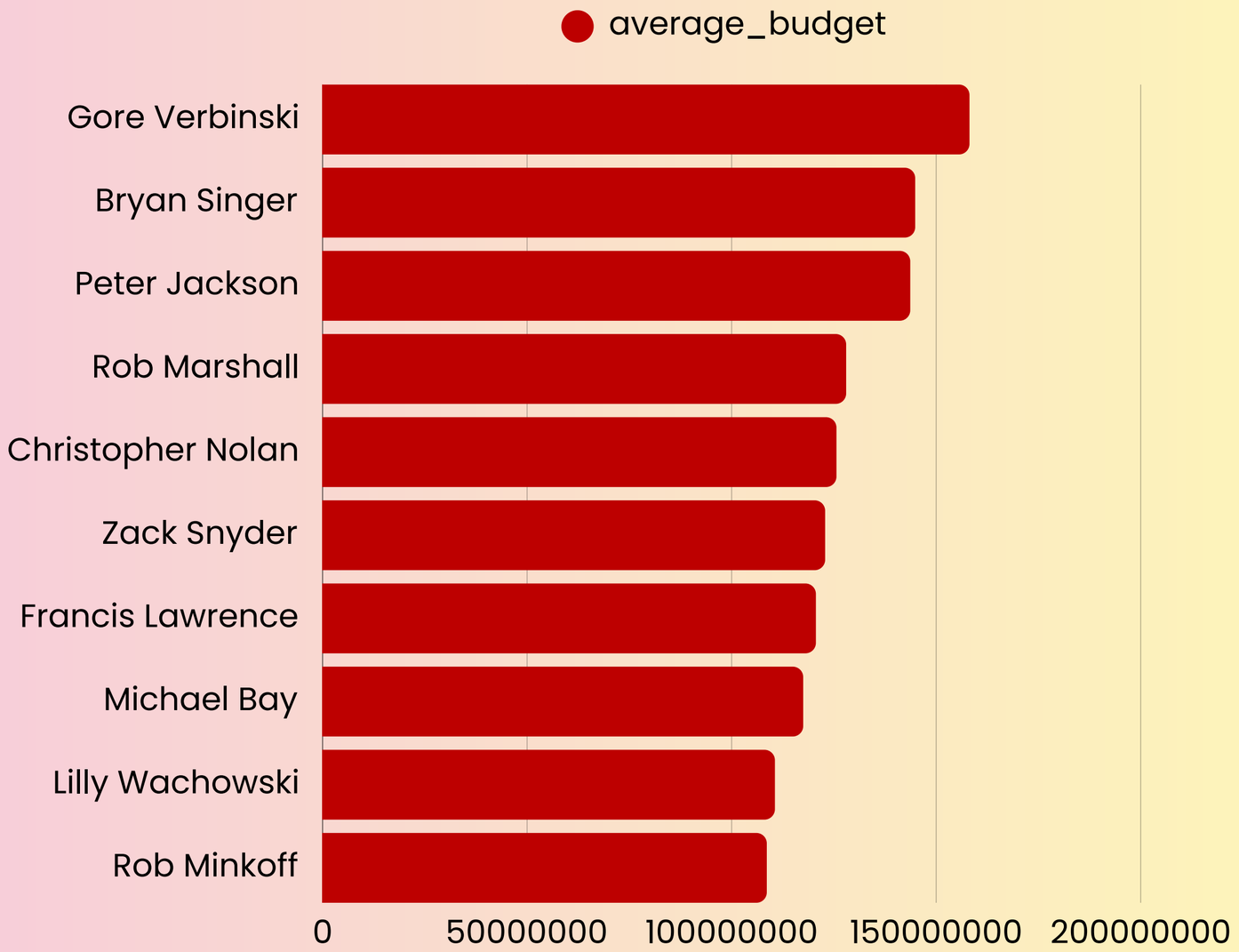
Exploratory Data Analysis

Methods

- Data visualization (scatter plots, histograms)
- Correlation analysis
- Grouping by categories (e.g., director)

Key Insights

- Strong positive relationship between budget and revenue
- Data is highly skewed, requiring log transformation
- Directors show variation in average budgets
- Weak relationship between runtime and ratings

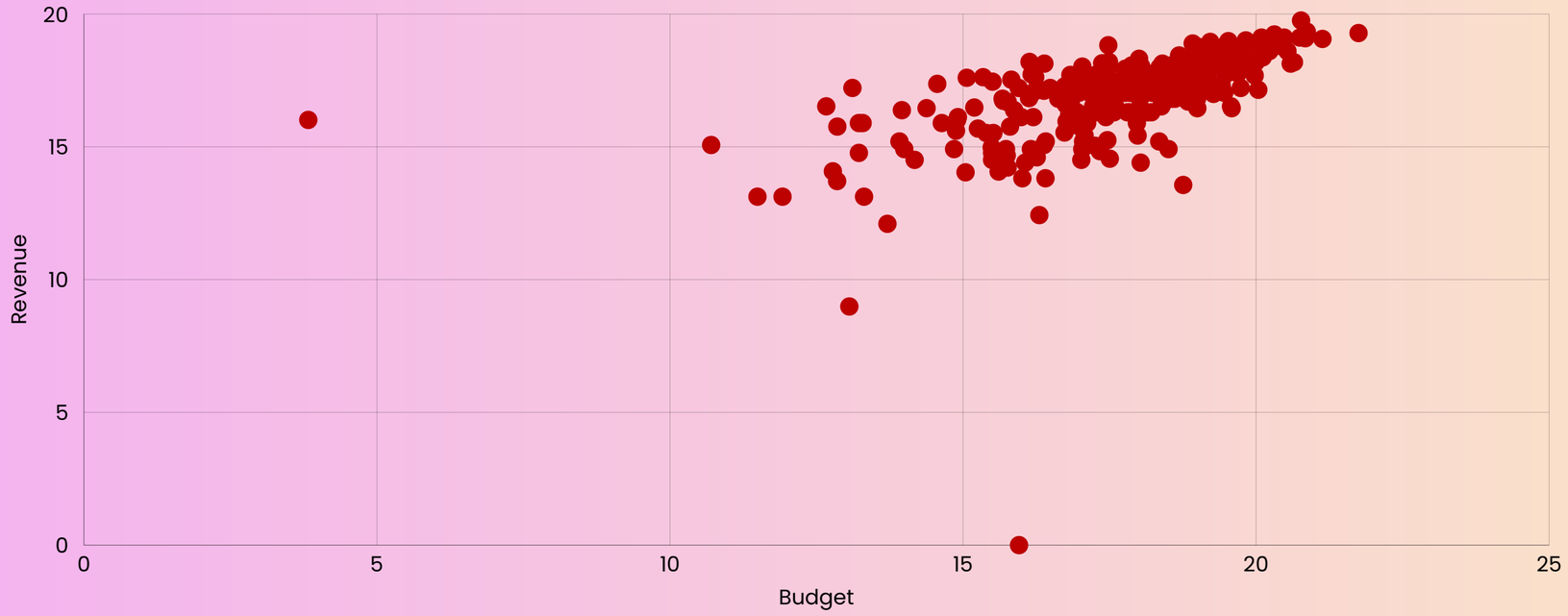


HYPOTHESIS 1

Films with larger production budgets generate higher box office revenue.

Finding & Analysis

- Moderate positive correlation between budget and revenue (≈ 0.66)
- Log transformation revealed a clearer relationship
- Films with higher budgets generally earn more



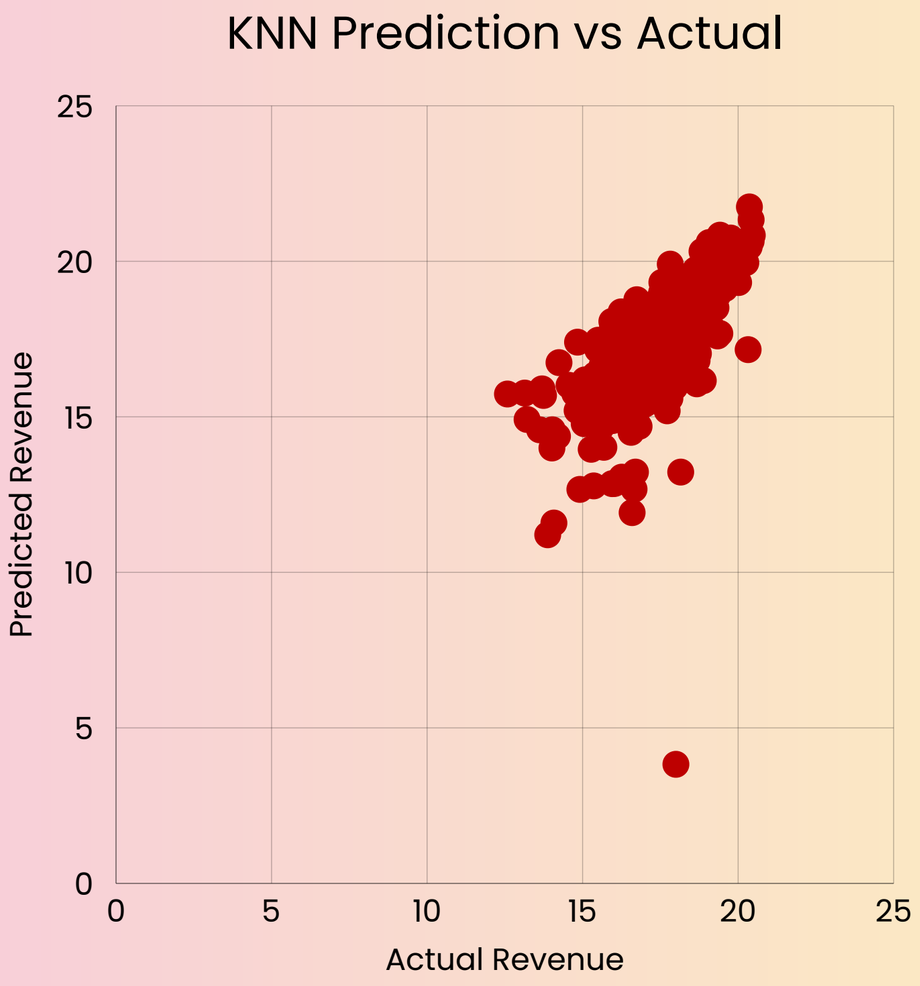
*We observed a clear positive trend, supporting the hypothesis

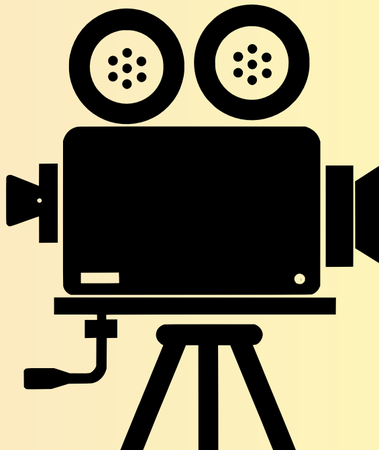
HYPOTHESIS 1

Films with larger production budgets generate higher box office revenue.

Machine Learning Results

- Models tested:
  - Linear Regression ($R^2 = 0.45$)
  - Decision Tree ($R^2 = 0.37$)
  - KNN ($R^2 = 0.53$)
- Best model: KNN
- Model insights:
  - Captures general trends
  - Less accurate for extreme values





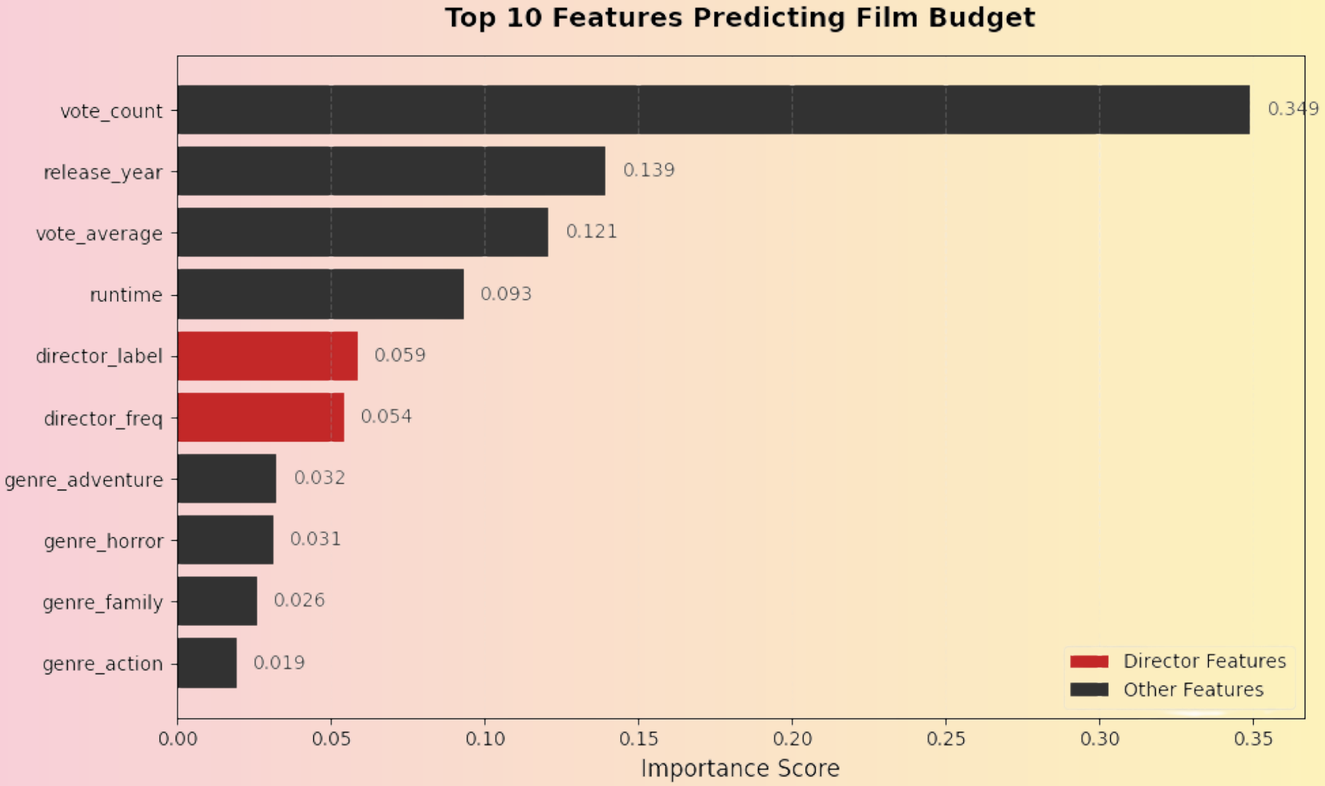
HYPOTHESIS 2

The choice of director significantly influences the production budget of a film.

Engineering: Encoded director identity (label) and experience (frequency)

Strong predictive signal: Director features rank 5th & 6th in overall importance

Key Insight: Director influence outperforms major genres (Action, Horror, Adventure)



HYPOTHESIS 2

The choice of director significantly influences the production budget of a film.

Machine Learning Results

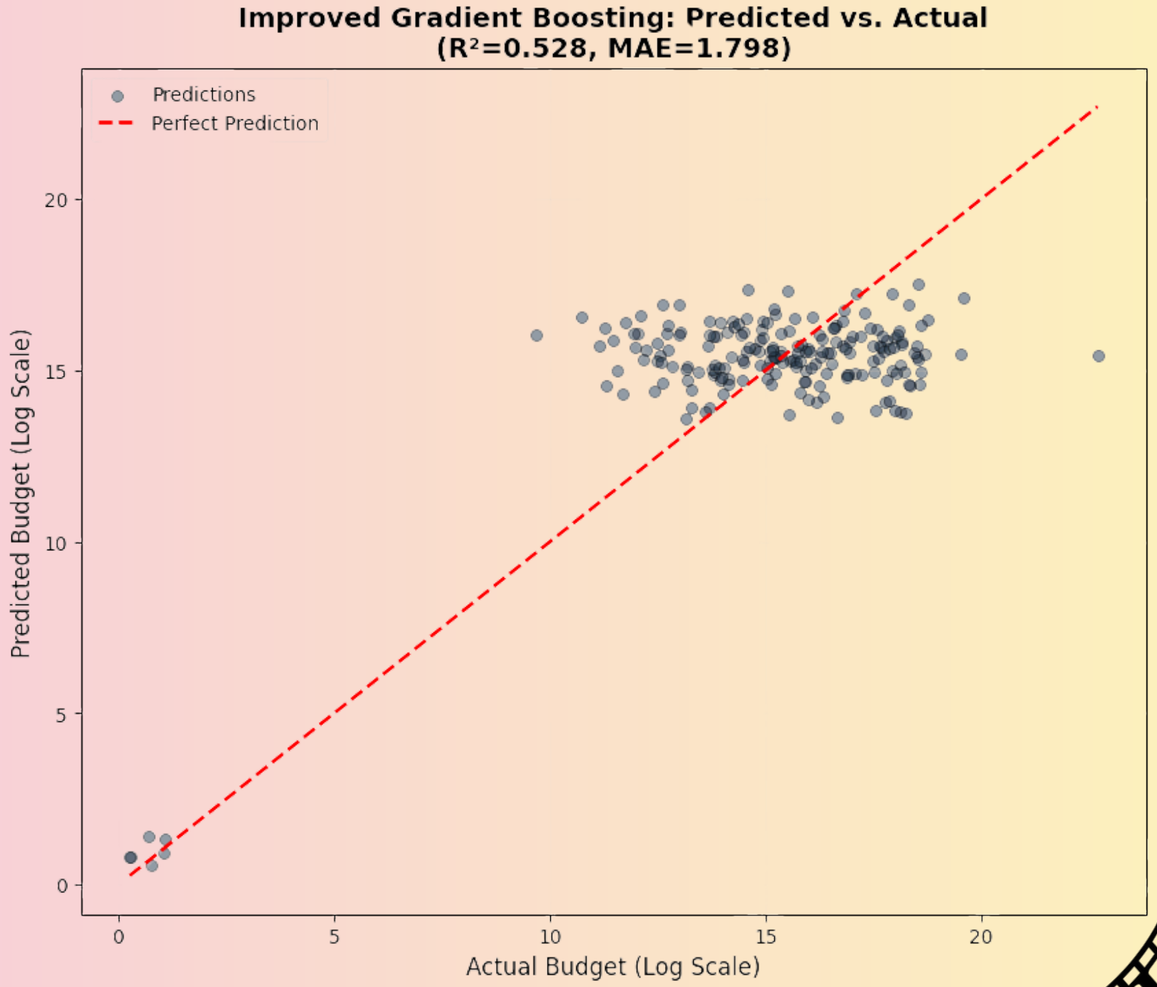
Models tested:

- Linear Regression (R^2 = 0.28)
- Random Forest (R^2 = 0.35)
- Gradient Boosting (R^2 = 0.53)

Best model: Tuned Gradient Boosting Regressor

Model insights:

- Performance: R^2 = 0.53
- Key Improvement: Fixed systematic overestimation of micro-budget films
- Accuracy: Average prediction error reduced to ~€1.8M (log scale MAE: 1.80)
- Robustness: Performs consistently across all budget scales (independent to blockbuster)



HYPOTHESIS 3

Films with longer runtimes tend to receive lower popularity or audience approval compared to shorter films

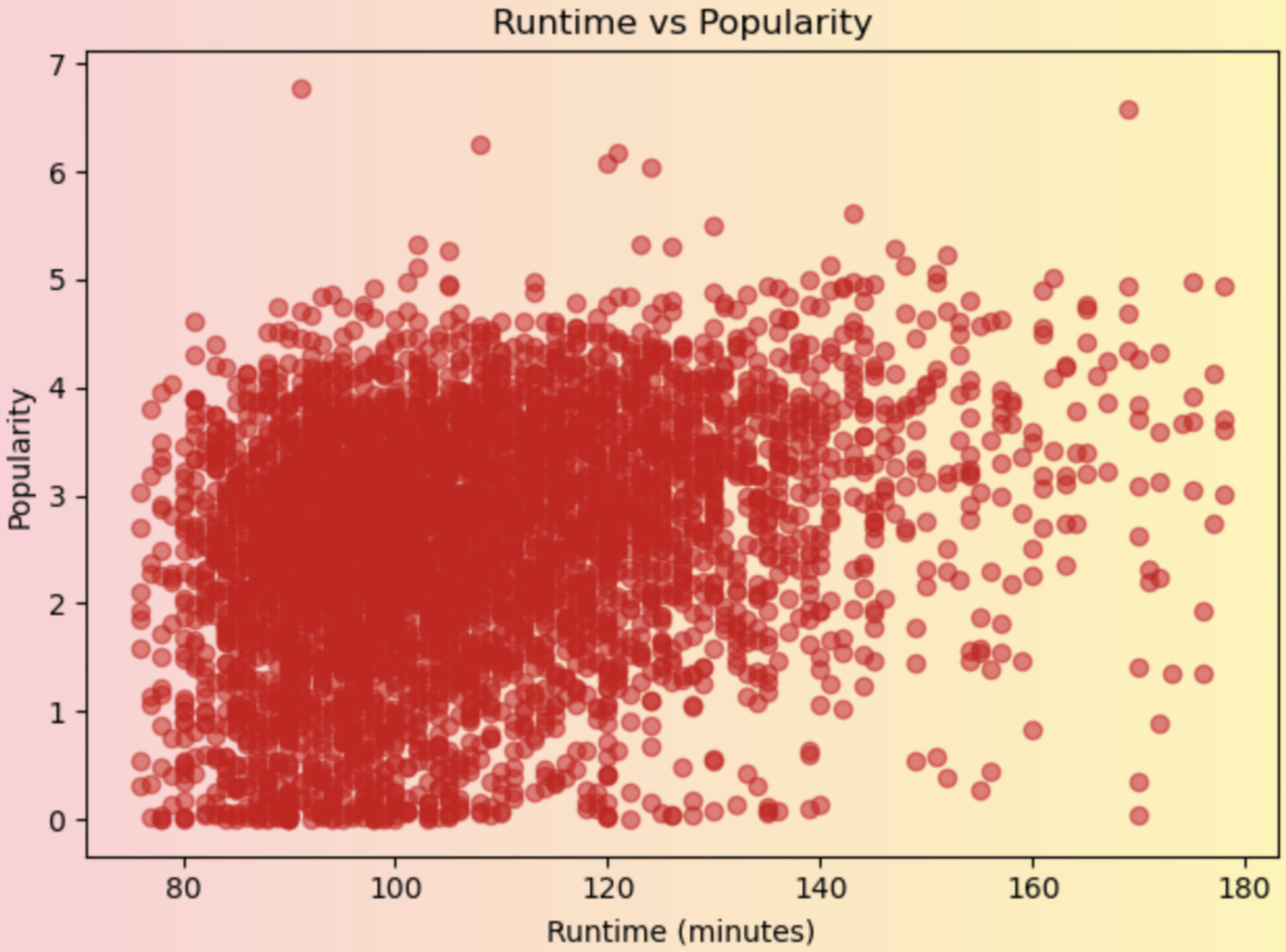
Objective:
Predict film popularity and test whether runtime influences audience reception.

Data:
~3,700 films
Features: Runtime, Genre, Budget, Popularity (target)

Finding & Analysis

- Weak relationship between runtime and popularity
- No clear trend in the data
- Longer films are not consistently less popular

We observed no strong pattern supporting the hypothesis



HYPOTHESIS 3

Films with longer runtimes tend to receive lower popularity or audience approval compared to shorter films

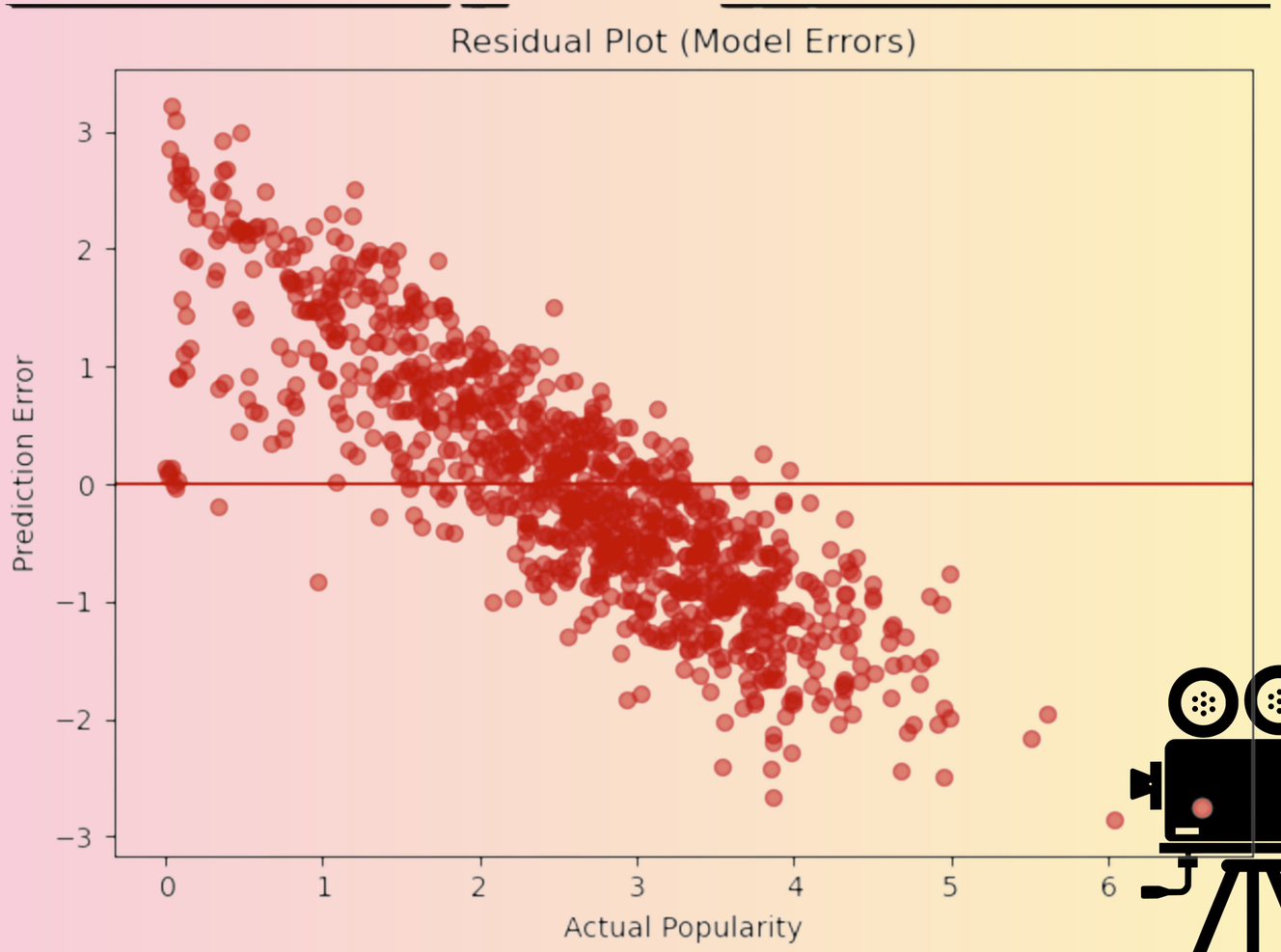
Machine Learning Results

Models tested:

- Linear Regression ($R^2 \approx 0.16$)
- KNN ($R^2 \approx 0.09$)
- Random Forest ($R^2 < 0$)

Best model: Linear Regression

Model errors show a clear pattern, meaning predictions are biased and not reliable.



CONCLUSIONS

HYPOTHESES RESULTS

H1: Budget → Revenue

- Supported
- Moderate positive relationship (~0.66)
- Higher budgets generally lead to higher revenue

H2: Director → Budget

- Partially supported
- Some directors consistently have higher budgets
- But variation exists across films

H3: Runtime → Ratings

- Weak or no clear relationship
- Runtime does not strongly impact audience approval

CONCLUSIONS

KEY INSIGHTS & LIMITATIONS

Key Learnings & Insights

- Budget is an important factor, but not the only driver of success
- Movie revenue is influenced by multiple factors
- Data required significant preprocessing (skewness, missing values)
- Machine learning models showed moderate predictive power ($R^2 \approx 0.53$)

Limitations

- Missing or incomplete data
- Limited variables (e.g., no marketing data)
- Some features simplified (genres, companies)

Future Improvements

- Include more features (cast, marketing, trends)
- Use more advanced models
- Improve feature engineering

THANK YOU